

SCIENCE

Science, Grade 12

University/College Preparation

SNC4M

This course enables students, including those pursuing postsecondary programs outside the sciences, to increase their understanding of science and contemporary social and environmental issues in health-related fields. Students will explore a variety of medical technologies, pathogens and disease, nutritional science, public health issues, and biotechnology. The course focuses on the theoretical aspects of the topics under study and helps refine students' scientific investigation skills.

Prerequisite: Science, Grade 10, Academic, or any Grade 11 university, university/college, or college preparation course in science

Big Ideas

Medical Technologies

- Medical technologies can have positive and negative effects on society, human health, the economy, and the environment.
- Knowledge of medical technologies, and the science behind them, can help patients better understand their diagnoses and treatment options.

Pathogens and Disease

- Appropriate technologies and informed choices with respect to personal behaviour can limit the spread of pathogens and diseases.
- Methods used to control the spread of pathogens and diseases can have both positive and negative effects on human health.

Nutritional Science

- The nutrients and other substances found in foods affect human health and well-being.
- An understanding of the role of nutrients and other substances found in foods enables people to make healthy lifestyle choices.

Science and Public Health Issues

- An understanding of threats to public health helps individuals and societies adopt appropriate practices to protect their health and the health of others.
- A global approach to public health is necessary to help prevent future pandemics.

Biotechnology

- Many social, ethical, and legal issues and conflicting interests have to be considered when determining the appropriate uses of biotechnology.
- Scientific knowledge helps individuals and society make informed decisions regarding biotechnology.

Fundamental Concepts Covered in This Course (see also page 5)

Fundamental Concepts	Medical Technologies	Pathogens and Disease	Nutritional Science	Science and Public Health Issues	Biotechnology
Matter	✓	✓	✓		✓
Energy	✓		✓		
Systems and Interactions		✓	✓	✓	
Structure and Function	✓	✓			
Sustainability and Stewardship			✓		✓
Change and Continuity		✓		✓	✓

A. SCIENTIFIC INVESTIGATION SKILLS AND CAREER EXPLORATION

OVERALL EXPECTATIONS

Throughout this course, students will:

- A1.** demonstrate scientific investigation skills (related to both inquiry and research) in the four areas of skills (initiating and planning, performing and recording, analysing and interpreting, and communicating);
- A2.** identify and describe careers related to the fields of science under study, and describe the contributions of scientists, including Canadians, to those fields.

SPECIFIC EXPECTATIONS

A1. Scientific Investigation Skills

Throughout this course, students will:

Initiating and Planning [IP]*

- A1.1** formulate relevant scientific questions about observed relationships, ideas, problems, or issues, make informed predictions, and/or formulate educated hypotheses to focus inquiries or research
- A1.2** select appropriate instruments (e.g., respirometer, titration apparatus) and materials (e.g., prepared slides, Petri dishes, food samples), and identify appropriate methods, techniques, and procedures, for each inquiry
- A1.3** identify and locate a variety of print and electronic sources that enable them to address research topics fully and appropriately
- A1.4** apply knowledge and understanding of safe laboratory practices and procedures when planning investigations by correctly interpreting Workplace Hazardous Materials Information System (WHMIS) symbols; by using appropriate techniques for handling and storing laboratory equipment and materials and disposing of laboratory materials, including biological waste (e.g., techniques to prevent contamination of specimens); and by using appropriate personal protection (e.g., wearing gloves when handling biological specimens)

Performing and Recording [PR]*

- A1.5** conduct inquiries, controlling relevant variables, adapting or extending procedures as required, and using appropriate materials and equipment safely, accurately, and effectively, to collect observations and data
- A1.6** compile accurate data from laboratory and other sources, and organize and record the data, using appropriate formats, including tables, flow charts, graphs, and/or diagrams
- A1.7** select, organize, and record relevant information on research topics from a variety of appropriate sources, including electronic, print, and/or human sources, using suitable formats and an accepted form of academic documentation

Analysing and Interpreting [AI]*

- A1.8** synthesize, analyse, interpret, and evaluate qualitative and/or quantitative data to determine whether the evidence supports or refutes the initial prediction or hypothesis and whether it is consistent with scientific theory; identify sources of bias and/or error; and suggest improvements to the inquiry to reduce the likelihood of error
- A1.9** analyse the information gathered from research sources for logic, accuracy, reliability, adequacy, and bias

* The abbreviation(s) for the broad area(s) of investigation skills – IP, PR, AI, and/or C – are provided in square brackets at the end of the expectations in strands B–F to which the particular area(s) relate (see pp. 20–22 for information on scientific investigation skills).

A1.10 draw conclusions based on inquiry results and research findings, and justify their conclusions with reference to scientific knowledge

Communicating [C]*

A1.11 communicate ideas, plans, procedures, results, and conclusions orally, in writing, and/or in electronic presentations, using appropriate language and a variety of formats (e.g., data tables, laboratory reports, presentations, debates, simulations, models)

A1.12 use appropriate numeric, symbolic, and graphic modes of representation, and appropriate units of measurement (e.g., SI and imperial units)

A1.13 express the results of any calculations involving data accurately and precisely, to the appropriate number of decimal places or significant figures

A2. Career Exploration

Throughout this course, students will:

A2.1 identify and describe a variety of careers related to the fields of science under study (e.g., nuclear medicine technician, nurse practitioner, hematologist, dietitian, geneticist) and the education and training necessary for these careers

A2.2 describe the contributions of scientists, including Canadians (e.g., Frederick Banting, John A. Hopps, Louis Siminovitch, Jean Cuthand Goodwill, Nancy Olivieri), to the field under study

B. MEDICAL TECHNOLOGIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** assess the impact of medical technologies and therapies, both conventional and alternative, used to diagnose and treat human health conditions;
- B2.** investigate the uses of, and analyse the information provided by, a variety of medical technologies;
- B3.** demonstrate an understanding of the function and use of a variety of medical technologies and the information they provide about the human body.

SPECIFIC EXPECTATIONS

B1. Relating Science to Technology, Society, and the Environment

By the end of this course, students will:

- B1.1** assess the costs and benefits of a conventional medical technology, therapy, or device that is used to diagnose or treat a human health condition (e.g., diagnostic technologies such as X-rays and ultrasound; surgical procedures such as laser removal of tumours; biomedical devices such as prosthetics) [AI, C]

Sample issue: Reproductive technologies have permitted many infertile couples to have babies. However, treatments can be expensive, and the long-term effects of strong hormone therapies that accompany in vitro fertilization are unknown. Moreover, some cultures routinely use these technologies for sex selection, resulting in skewed sex ratios and accompanying social problems.

Sample questions: What are the benefits of using nuclear isotopes in the diagnosis of disease? What are the costs of this technology, including the potential long-term health effects for patients? What are the benefits of organ transplants? Why is there a black market in human organs, and what can be done to stop it? What are the benefits of telesurgery for people living in remote areas?

- B1.2** identify a variety of alternative technologies and therapies used to diagnose or treat human health conditions (e.g., biofeedback, acupuncture, homeopathy, chiropractic, Aboriginal healing practices), and assess the effectiveness of one such therapy [AI, C]

Sample issue: Naturopathic medicine focuses on the ability of the body to heal and maintain itself. Practitioners emphasize a holistic approach to health and use herbal medicines and non-invasive treatments. However, there is a lot of controversy over the effectiveness of some herbal treatments and proposals for their regulation.

Sample questions: How effective are botanical treatments for skin irritations? What are their limitations? How does acupuncture work? What sort of ailments is it used to treat? How effective is it? What is biofeedback? What sorts of conditions does it treat effectively? What are its limitations?

B2. Developing Skills of Investigation and Communication

By the end of this course, students will:

- B2.1** use appropriate terminology related to medical technologies, including, but not limited to: *aseptic techniques, feedback loop, biochemical, and biomechanical* [C]
- B2.2** use a variety of medical technologies to collect data related to blood pressure, heart rate, lung capacity, and body mass, and analyse the data (e.g., use a stethoscope to determine heart rate under various conditions; use a respirometer to measure lung capacity) [PR, AI]
- B2.3** use a microscope to investigate prepared slides of human blood, identifying blood cells by type, size, and number, and determine the ratios between different cells in the samples [PR, AI]

B2.4 interpret information generated by a variety of imaging technologies (e.g., X-rays, ultrasound), and communicate their findings [AI, C]

B3. Understanding Basic Concepts

By the end of this course, students will:

B3.1 explain the four primary vital signs in humans (i.e., body temperature, heart rate, blood pressure, respiration rate)

B3.2 describe the normal range for various physiological and biochemical indicators (e.g., heart rate, lung capacity, blood pressure, blood sugar)

B3.3 explain the function and use of a variety of medical devices and technologies for diagnostic and treatment purposes (e.g., sphygmomanometer, stethoscope, ultrasound, X-ray, computerized axial tomography [CAT] scan, pacemaker, chemotherapy)

B3.4 describe the function and use of technologies, devices, and techniques for biomedical repair (e.g., prosthetics, artificial organs, plastic surgery)

B3.5 describe a recent technological development or advance in diagnosis or treatment in the health care field (e.g., artificial skin for burn victims, artificial and transgenic organ transplants, smart drugs, nanotechnologies, biophotonics)

C. PATHOGENS AND DISEASE

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** evaluate the impact of scientific and technological knowledge and individual behaviour on the control of pathogens and the prevention of disease;
- C2.** investigate the nature and growth of pathogens and the effectiveness of measures intended to prevent their spread;
- C3.** demonstrate an understanding of pathogens, the diseases they cause, and ways of controlling their spread.

SPECIFIC EXPECTATIONS

C1. Relating Science to Technology, Society, and the Environment

By the end of this course, students will:

- C1.1** analyse, on the basis of research, the impact, both positive and negative, of scientific and technological advances intended to prevent the spread of illness and disease [IP, PR, AI, C]

Sample issue: The development of antibiotics such as penicillin and sulfa in the twentieth century saved many lives. However, overuse and improper use of antibiotics have contributed to the development of antibiotic-resistant forms of bacteria that threaten public health.

Sample questions: How does the irradiation of food reduce the incidence of food-borne illness? Why is the use of this technology controversial? What are the pros and cons of using insecticides such as DDT to kill malaria-carrying mosquitoes? What are the advantages and disadvantages of using “no touch” technologies to stem the spread of diseases?

- C1.2** evaluate the impact of individual choices (e.g., with respect to vaccination, the proper use of antibiotics or mosquito repellent) on the control of pathogens and the prevention of disease [AI, C]

Sample issue: Intravenous drug users who share needles risk transmitting or contracting serious diseases such as hepatitis and HIV/AIDS. Programs such as needle exchanges and safe

injections sites reduce the risks posed by shared needle use, but their success depends on the initiative of the drug users.

Sample questions: What measures should food handlers take to prevent the spread of disease? What precautions should travellers take in order not to spread disease? Why is it important to stay home when you have a communicable illness?

C2. Developing Skills of Investigation and Communication

By the end of this course, students will:

- C2.1** use appropriate terminology related to pathogens and diseases, including, but not limited to: *parasite*, *epidemiology*, *pathogenesis*, and *vector* [C]
- C2.2** analyse, on the basis of inquiry, the effects of various treatments on pathogenesis (e.g., the effect of mouthwash or penicillin on the growth of bacteria) [PR, AI]
- C2.3** analyse, using prepared slides or computer simulations, the characteristics, properties, and virulence of various bacteria [PR, AI]
- C2.4** use an inquiry process to demonstrate the effect of the use of sterile techniques (e.g., pasteurization, use of an autoclave) on pathogenesis [IP, PR]

C3. Understanding Basic Concepts

By the end of this course, students will:

- C3.1** describe the characteristics and life cycles, including reproductive cycles, of representative pathogens (e.g., lysogenic cycle; lytic cycle; infectious cycle of malaria)
- C3.2** describe the mode of transmission of various diseases, including those that are insect-borne (e.g., malaria, encephalitis), airborne (e.g., influenza, tuberculosis), water-borne (e.g., cholera, poliomyelitis), sexually transmitted (e.g., HIV/AIDS), and food-borne (e.g., mad cow disease, trichinosis, salmonella)
- C3.3** explain how the human immune response acts as a natural defence against infection
- C3.4** describe the role of vaccines, antibiotics, antiretrovirals, and other drug therapies and antiseptics in the control of pathogenesis
- C3.5** describe non-medical ways to protect oneself from contracting pathogenic disease in a variety of situations (e.g., aseptic techniques such as wearing sterile gloves; proper personal hygiene such as frequent and thorough hand washing; the use of insect repellent)
- C3.6** describe some of the means used by international non-governmental organizations (e.g., Médecins sans Frontières, Oxfam, Ryan's Well Foundation, UN agencies, the Stephen Lewis Foundation) to control the spread of disease (e.g., distribution of vaccines, medication, malaria nets; installing wells so people have access to clean water; public education on strategies for transmission prevention)
- C3.7** describe aseptic techniques used in the workplace, and explain their importance in preventing the spread of pathogens (e.g., cooking meat to a safe temperature and refrigerating leftovers quickly to avoid growth of bacteria in restaurant food; frequent hand sanitizing and use of sterile gloves in hospitals to prevent the spread of pathogens to vulnerable populations)

D. NUTRITIONAL SCIENCE

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** assess how personal and societal factors affect eating behaviours, and evaluate the social and economic impact of the use of non-nutrient food additives;
- D2.** investigate chemical components of and energy in food, and the processes by which food is digested;
- D3.** demonstrate an understanding of chemical components of and energy in food, and the processes by which food is digested.

SPECIFIC EXPECTATIONS

D1. Relating Science to Technology, Society, and the Environment

By the end of this course, students will:

- D1.1** analyse the social and economic costs and benefits of the use of non-nutrient food additives in food preservation and food enhancement techniques (e.g., sulfites in dried fruit; food colouring; MSG) [AI, C]

Sample issue: Non-nutrient food additives increase the shelf life of many foods, decreasing waste from spoiled food and allowing foods to be transported around the globe. Importing food increases consumer choice, yet it also increases carbon emissions and can hurt the Canadian farm economy.

Sample questions: What types of additives are used to make vegetables in salad bars and grocery stores stay green and fresh looking? How do these additives benefit the retailer? What effect do they have on consumers? Why are nitrates added to many types of processed meat? What sorts of foodstuffs contain artificial colour? What health risks are associated with these additives?

- D1.2** evaluate the impact of some personal and societal factors (e.g., allergies, disease, body image, cultural preferences) on eating behaviours (e.g., the relationship between societal ideals of beauty and interest in “fad” diets) [AI, C]

Sample issue: A nutritious diet is an important component of a healthy lifestyle. Yet, decisions about what to eat can be influenced by social factors such as the availability of fast food in a fast-paced society, peer pressure, and the

pervasive images presented by the fashion and entertainment industries.

Sample questions: What effect does advertising have on your food choices? How do food allergies affect the diets of individuals and the food choices available in school cafeterias? Why do people with diabetes have to monitor their consumption of sugar and carbohydrates? What impact does anorexia have on a person’s eating behaviours? What effect can a person’s cultural background have on their food choices?

D2. Developing Skills of Investigation and Communication

By the end of this course, students will:

- D2.1** use appropriate terminology related to nutritional science, including, but not limited to: *macromolecules, protein, starch, vitamin, carbohydrate, fats, lipids, pepsin, and amylase* [C]
- D2.2** plan and conduct chemical tests on a variety of foods to determine their chemical components (e.g., protein, starch, fats, lipids, carbohydrates, vitamins) [IP, PR, AI]
- D2.3** investigate how enzymes break down macromolecules (e.g., amylase digests starch; pepsin and hydrochloric acid digest protein), and test the products of different types of digestion (e.g., use Benedict’s solution to test for the presence of simple sugars produced by the digestion of carbohydrates) [PR]
- D2.4** investigate the process of emulsification in fats and lipids (e.g., using commercially obtained bile and cooking oil) [PR]

D2.5 conduct titrations to determine the effects of various antacids on hydrochloric acid [PR, AI]

D2.6 plan and conduct an inquiry to determine the nutrient or energy content in selected food samples (e.g., hamburger, bread) [IP, PR, AI]

D3. Understanding Basic Concepts

By the end of this course, students will:

D3.1 describe the basic chemical components of proteins, carbohydrates, fats and lipids, and vitamins and minerals, and explain their functions in the body

D3.2 explain how laboratory methods are used to determine the relative energy content of food (e.g., use of a calorimeter)

D3.3 describe requirements for a balanced diet based on the biochemical and energy needs of the average body, and explain how these requirements might vary among people with different lifestyles (e.g., young children, the elderly, a person with diabetes, an athlete)

D3.4 describe the structure and function of the components of the digestive system (e.g., mouth, tongue, epiglottis, esophagus, stomach, intestines, liver, gall bladder, pancreas, appendix, rectum, anus, salivary glands, saliva, bile) with respect to physical and chemical digestion

D3.5 describe optimum conditions for the effective functioning of some digestive enzymes found within the human body (e.g., amylase, pepsin)

E. SCIENCE AND PUBLIC HEALTH ISSUES

OVERALL EXPECTATIONS

By the end of this course, students will:

- E1.** assess the impact of scientific research, technological advances, and government initiatives on public health;
- E2.** investigate various strategies related to contemporary public health issues;
- E3.** demonstrate an understanding of major public health issues, past and present.

SPECIFIC EXPECTATIONS

E1. Relating Science to Technology, Society, and the Environment

By the end of this course, students will:

- E1.1** assess the impact of scientific research and technological advances on public health around the world (e.g., widespread immunization for diseases such as polio, telemedicine for people in remote areas, new drug therapies to combat disease) [AI, C]

Sample issue: In developed countries, people living with HIV/AIDS have access to highly effective but expensive drugs, which most people in the developing world cannot afford. Although patent restrictions have prevented the widespread use of cheaper generic drugs, some of these restrictions have recently been relaxed, which may decrease the global death toll from AIDS.

Sample questions: How are monitoring and reporting technologies used to contain the spread of infectious diseases such as avian flu and severe acute respiratory syndrome (SARS)? How effective is the use of mosquito nets in reducing the spread of malaria?

- E1.2** assess, on the basis of research, the effectiveness of a municipal, provincial, or federal government initiative intended to protect the public health of Canadians (e.g., immunization programs, smoking bans, Health Canada advisories) [IP, PR, AI, C]

Sample issue: The Canadian Food Inspection Agency is responsible for administering laws

that regulate agriculture and the food-processing industry. Its mandate includes monitoring farm animals for avian flu and mad cow disease and ensuring the safety of food additives. Although agency regulations have made food safer, each year many Canadians contract food-borne illnesses.

Sample questions: How effective have the safety measures for blood donation and transfusions been in reducing the spread of blood-borne diseases since the tainted-blood scandal of the 1980s? What is the impact on immunization programs of claims that vaccinations can cause severe side effects? How useful is the information on emergency preparedness and response in your local area?

E2. Developing Skills of Investigation and Communication

By the end of this course, students will:

- E2.1** use appropriate terminology related to public health issues, including, but not limited to: *pandemic*, *contamination*, *infectious disease*, *quarantine*, and *vaccination* [C]
- E2.2** analyse and interpret, using a case study or research data, scientific evidence regarding the effectiveness of a public health program intended to reduce disease transmission (e.g., distribution of bed nets to fight malaria; safe injection sites for intravenous drug users; programs to encourage hand washing in hospitals to stop the spread of *C. difficile*) [AI, C]

E2.3 use a research process to investigate public health strategies developed to combat a potential pandemic (e.g., SARS, C. difficile, avian flu) [IP, PR]

E2.4 use a research process to locate a media report on a public health issue (e.g., the handling of SARS, the banning of bisphenol-A in plastic bottles), summarize its arguments, and assess them from a scientific perspective [IP, PR, AI, C]

E3. Understanding Basic Concepts

By the end of this course, students will:

E3.1 describe the characteristics according to which a pandemic is classified (e.g., the strain of a virus, its mode of transmission)

E3.2 explain how pandemics have affected humanity throughout history (e.g., the bubonic plague of 1347–1352 in Europe, the cholera pandemic of 1817–1823 in Asia, the global Spanish influenza pandemic of 1918–1920, the contemporary AIDS pandemic)

E3.3 explain the impact of various threats to public health, including infectious diseases (e.g., hepatitis, HIV/AIDS, tuberculosis, malaria, sexually transmitted diseases), chronic diseases (e.g., cardiovascular disease, diabetes, asthma), and environmental factors (e.g., climate change, air pollution, chemical pollutants, radiation)

E3.4 explain a variety of social factors that can promote the rapid spread of infectious diseases (e.g., global population growth, international travel, poor sanitation, lack of clean drinking water)

E3.5 describe public health measures, including legislation, that are used for the protection of the public (e.g., quarantines, vaccinations, water chlorination, regulations on what items travellers can bring into a country)

E3.6 explain why some populations are particularly susceptible to specific health problems (e.g., the risk of diabetes among First Nations populations; the risk of thalassemia among Mediterranean populations; the risk of pneumonia and tuberculosis among people with HIV/AIDS)

F. BIOTECHNOLOGY

OVERALL EXPECTATIONS

By the end of this course, students will:

- F1.** analyse a variety of social, ethical, and legal issues related to applications of biotechnology in the health, agricultural, or environmental sector;
- F2.** investigate various techniques used in biotechnology and how they are applied in the food industry and the health and agricultural sectors;
- F3.** demonstrate an understanding of biological processes related to biotechnology and of applications of biotechnology in the health, agricultural, and environmental sectors.

SPECIFIC EXPECTATIONS

F1. Relating Science to Technology, Society, and the Environment

By the end of this course, students will:

- F1.1** analyse social issues related to an application of biotechnology in the health, agricultural, or environmental sector (e.g., issues related to the uses of genetically modified organisms or to the uses and availability of in vitro fertilization) [AI, C]

Sample issue: The promise of genetically modified (GM) crops was that they would be resistant to pests and would produce more abundant harvests. However, GM crops can crossbreed with crops in adjoining fields, thus contaminating traditional food sources, reducing biodiversity, changing farming practices, and limiting the choices available to consumers.

Sample questions: What was the social impact of the discovery of insulin? What is the potential impact on the family of biotechnological developments related to the treatment of infertility and the genetic screening of embryos? What are the social benefits of the development of drought-resistant crops?

- F1.2** analyse, on the basis of research, ethical and legal issues related to an application of biotechnology in the health, agricultural, or environmental sector (e.g., ethical questions related to xenotransplantation; legal issues related to access to an individual's genetic information) [IP, PR, AI, C]

Sample issue: Advances in biotechnological research in human health have raised many difficult questions. Currently, scientists, ethicists, politicians, and business people are debating issues such as who should have access to a person's genetic information, whether human cloning should be permitted, and whether human embryos should be used for stem-cell research.

Sample questions: If a disease has no known cure, should we use biotechnology to predict or diagnose its occurrence in individuals? Why or why not? Who owns or controls frozen embryos left over after in vitro fertilization? Who determines whether genetically modified foods are safe? How might the testing/regulation process be open to abuse? What are the legal and ethical implications of introducing into an ecosystem a species engineered through biotechnology?

F2. Developing Skills of Investigation and Communication

By the end of this course, students will:

- F2.1** use appropriate terminology related to biotechnology, including, but not limited to: *selective breeding, hybridization, replication, mutation, genomics, and gene therapy* [C]
- F2.2** plan and conduct an inquiry into various traditional biotechnological techniques used in the food industry (e.g., the use of fermentation to produce bread, cheese, yogurt) [IP, PR]

F2.3 investigate, through laboratory inquiry or computer simulation, a recently developed biotechnological method used in the health sector (e.g., the process of electrophoresis to degrade DNA) [PR]

F2.4 investigate, through laboratory inquiry or computer simulation, a recently developed biotechnological method used in the field of agriculture (e.g., bioremediation of a chemical fertilizer spill; the cloning of corn; the use of synthetic hormones to promote growth in live-stock) [PR]

F3. Understanding Basic Concepts

By the end of this course, students will:

F3.1 explain various methods used, over time, in the field of biotechnology (e.g., use of living organisms to make or modify products, selective breeding to create particular breeds of animals, manipulation of genes to develop organisms with particular traits)

F3.2 explain the structure and functions of macromolecules (e.g., DNA, RNA) and the synthesis of proteins (e.g., transcription, translation, gene expression)

F3.3 describe applications of biotechnology in the health (e.g., genomics, gene therapy, xenotransplantation, in vitro fertilization), agricultural (e.g., genetically modified crops, biopesticides, cloning), and environmental sectors (e.g., bioremediation, phytoremediation)